A Depth-Agile Vertical Array for Bidirectional Acoustic Data Recovery from Ocean-Bottom Seismometers

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A preliminary version of this work has been submitted to a conference. This manuscript extends it with a fuller treatment of the propagation method, the park-depth analysis of Section VII-A, an expanded limitations assessment, and additional figures.

Abstract

Ocean-bottom seismometers record for a year or more, but their data is unreachable until a vessel recovers the instrument. Acoustically linked surface moorings solve this within a few kilometres, and their reach is limited by an effect transmit power cannot address: a signal from a seafloor node reaches the relay twice, directly and by surface echo, and the copies interfere. At a reference site these arrivals are 9.5° to 19.8° apart across 1–10 km, and a deployed moored observatory's 60° transducer suppresses the reflection by under 3 dB. A 24-element vertical line array with a 4.23° beamwidth suppresses it by 13.6 to 32.9 dB, varying by less than 0.3 dB between seasons. The array must be decoupled from wave motion, so it rides a buoyancy capsule that selects its park depth each dive from an onboard sound-speed cast. Propagation uses the Bellhop Gaussian beam tracer on measured oceanography: World Ocean Atlas 2023 profiles, GEBCO 2026 bathymetry, a 34-year wave climate from station C46206, and Hamilton's seabed values. The usable radius extends from 3.22 to 5.22 km in summer and 4.51 to 5.21 km in winter, and the same payload costs 62.4% less energy at that edge, independent of modem efficiency. An ablation confirms the array alone, at fixed depth, reaches only 3.07 km in winter.

Keywords: underwater acoustic communication; vertical line array; beamforming; multipath rejection; ocean-bottom seismometer; delay-tolerant network; buoyancy-driven platform; magnetoelectric antenna.

I. Introduction

A broadband ocean-bottom seismometer (OBS) placed on the deep seafloor writes continuously for about a year. None of that record can be read until a research vessel returns to the site, fires an acoustic release and waits for the instrument to float to the surface. Until that moment the science is inaccessible, instrument failures go undetected, and the sampling strategy cannot be adapted in response to events. For a seismic network intended to capture transient activity, this is a substantial operational constraint.

Frye *et al.* addressed the problem with an acoustically linked moored observatory deployed on the Nootka fault in 2004 [1]. A surface buoy carrying an acoustic modem and an Iridium terminal relayed data from two seafloor instruments for thirteen months at 5,400 bps, transferring more than 500 kB per day. The system worked, and it remains the reference architecture for this class of problem.

Its principal limitation is reach. The paper reports approximately 4 km of acoustic telemetry and describes the installation as serving “sensors located over a 5–20 square kilometer area” [1]. An instrument outside that circle is not served slowly; it is not served at all. The relay radius therefore determines which instruments are present in the network, rather than how quickly any individual link operates. Extending it is not a throughput question but a coverage one.

A second constraint is worth recording because it motivates the mechanical architecture below. The buoy’s transducers hang beneath a wave-driven float on a conductor bundle. That bundle failed within a month of deployment. The authors report that “the high-rate data stopped being received by the buoy” and traced the fault to “the analog receive path from the transducer to the DSP” [1]. Operation was restored in late September from a May

deployment, and the system thereafter ran on two channels rather than four. The failure is instructive: the authors had anticipated it, fitting a urethane composite “to protect the conductors from bending strain produced by the motion of the buoy in severe sea states” [1], and it occurred regardless. We do not quantify the reliability of the alternative proposed here, and make no claim about comparative availability; we note only that an architecture without an external conductor path does not possess this particular failure mode.

This paper presents an array architecture that addresses the reach limitation, together with a buoyancy capsule as one embodiment of it. The contributions are array-side and therefore platform-independent, applying equally to a moored relay, a glider, or an autonomous vehicle harvesting a sensor field.

First, we identify and measure the mechanism that sets the radius. The direct path and its surface reflection arrive between 9.5° and 19.8° apart in elevation across the operating range. A 60° transducer cannot separate them, because both arrivals fall within its main lobe. A 4.23° vertical array suppresses the reflection by 13.6 to 32.9 dB.

Second, we report a coverage result computed on measured oceanography. The usable radius rises from 3.22 km to 5.22 km in summer and from 4.51 km to 5.21 km in winter. The depth-agile radius varies by 0.2% between seasons against 40% for the moored baseline.

Third, we show that the resulting energy saving is independent of modem efficiency. The efficiency term cancels in the ratio of delivered-energy costs, so the fractional saving depends only on the two packet error rates.

Fourth, we characterise the park-depth window within which such an array can operate, bounded below by sea state and above by geometric reach, and show that the upper bound moves with target range so that no single fixed depth is simultaneously optimal.

We are explicit about what is not claimed. We do not claim that a single relay covers an arbitrary instrument field, nor that a drifting platform substitutes for a propelled vehicle where instruments are widely scattered. The claim is that the array hears better, and by reciprocity speaks better, at a given geometry, and that this benefit transfers to whatever

platform carries it. Throughout, quantities are tagged by provenance: [V] verified from a publication or public dataset, [C] computed from verified inputs, [S] simulated in this work, [D] a design choice, and [A] an assumption. All assumptions are enumerated in Section VIII.

II. Related Work

A. The site-matched baseline

Frye *et al.* [1] is the primary comparison and is uniquely appropriate for it: the same fault, the same water depth, a published operating range and published delivered data. We model its receiver from its own stated hardware rather than from assumption. The paper reports "directional transducers with 60 degree beams," a source level of 190 dB, approximately 8 W of transmit power, and transducers mounted at the bottom of a 10 m chain below the mooring's universal joint [1]. A 60° full-width cone subtends 0.842 sr, giving a directivity index of

$$DI = 10 \log_{10}(4\pi / 0.842) = 11.74 \text{ dB} \quad [C]$$

The baseline is therefore not omnidirectional. Any comparison that models it as such would overstate the advantage attributable to an array, and we avoid that error deliberately.

B. The mobile benchmark

Send, Regier and Jones [2] operate underwater gliders carrying acoustic modems to retrieve data from subsurface instruments in the California Current. Their site differs from ours, at 3,800–4,500 m water depth with a different sound-speed structure, so we use their results as a hardware and operational benchmark rather than as a head-to-head comparison: a measured 75% delivery rate, equivalent to 1.33 transmissions per delivered packet, and 0.49 J per delivered byte [2].

Two of their findings bear directly on the present work. Their Fig. 1 shows ray paths from transducers at 1,000 m and at 3,200 m, and notes a near-surface shadow zone beyond 6

km in the first case and none in the second. This is published evidence that source depth determines whether a shadow zone exists, which is the physical basis for depth agility. Their conclusion states that “in retrospect it might have been wiser to use a more directional transducer… as the increased directivity would increase the sound level received by the glider” [2]. The present work can be read as a quantification of that recommendation.

C. Link budget and error model

We adopt the link budget and Rayleigh error model of Khan and Cho [3] without modification, including their ambient-noise law, the BPSK error rate under Rayleigh fading, and a 1,024-bit data packet. Khan and Cho state explicitly that their sensor nodes carry omnidirectional transducers and that the directivity index is therefore taken as zero [3]. Introducing a non-zero directivity at the relay, inside their own equations and leaving every other term unchanged, is the intervention evaluated in this paper.

D. Reconfigurable acoustic surfaces

Zhao *et al.* [4] establish that acoustic shadow zones constitute a binary failure mode, with coverage below 20% regardless of radiated energy, and demonstrate a reconfigurable acoustic surface that restores it. Later work has carried acoustic surfaces from principle to tank-scale practice [19] and added in-phase and quadrature modulation [20]. We adopt their array gain law, $\mathrm{DI} = 10 \log_{10}(N)$, which they validate experimentally at $N = 2$ with a measured 2.69 dB against 3.01 dB predicted [4].

We cite this body of work for physics but do not simulate it as a baseline, for a structural reason. Their surface is a mid-path reflector: source, surface, destination, three points, with the surface depth required to satisfy both legs simultaneously. The capsule evaluated here is a store-and-forward terminal: two points, one hop, with the satellite link occurring hours later at the surface. Their optimisation acts on a middle position that this architecture does not possess. Section VII-B presents a fixed-depth versus depth-agile

comparison that serves the same purpose within our own geometry.

E. Mobile data collection and path planning

A parallel line of work collects from seafloor instruments using mobile vehicles, following the data-mule architecture of Shah *et al.* [21]. Hollinger *et al.* plan autonomous vehicle tours under a probabilistic communication model of the form $P(x_v, x_n) = f(D(x_v, x_n))$, in which the probability of successful collection decreases monotonically with separation [15]. Khan *et al.* extend this with clustered path planning and a Markov decision process [16], and multihop variants relay between instruments to shorten vehicle transits [22].

These works address where the receiver should go. The present work addresses how well it hears when it arrives. The two compose rather than compete. Hollinger *et al.* observe that as communication quality degrades their problem approaches the classical travelling-salesperson problem, because the vehicle must then approach each sensor closely [15]. The function f therefore governs how near a vehicle has to get, and in their formulation the usable radius reported in Section VI-B is the probabilistic neighbourhood around each instrument, which those methods take as a fixed input. We have not quantified the resulting change in tour length and make no claim about it. We note only that the vehicles in both works carry omnidirectional modems with no beamforming, multipath rejection or depth selection, so the receiver evaluated here is applicable to them unchanged.

F. The wider underwater networking context

Underwater wireless networking more broadly has been surveyed in [34]–[36]. Routing, localisation and energy management under acoustic constraints are each long-standing problems [37]–[41]. Optical and magnetic-induction links offer higher bandwidth over shorter ranges and are studied as complements to acoustics rather than replacements for it [42]–[46]. Machine-learning methods have been applied across the stack, from channel estimation to network control [47], [48]. On the mobile side, vehicles have been used for

hybrid optical–acoustic harvesting, energy-aware routing and multi-vehicle coordination [49]–[52], supported by acoustic positioning [53], and reconfigurable acoustic surfaces have been proposed for long-range and deep-sea deployment [54], [55]. Crossing the air–water interface directly, rather than relaying through a surfacing platform, has been demonstrated by translational acoustic-to-radio conversion, thermoacoustic coupling and backscatter techniques [56]–[59], though all three presently remain short-range.

III. System Description

A. Operating concept

The capsule is designed as a delay-tolerant, store-and-forward relay [23]. It is not intended for tsunami or earthquake early warning. At the surface, it receives its tasking before diving to a park depth determined from an onboard sound-speed cast. Once at depth, it beamforms on a seafloor node and buffers the received data. It then ascends and uplinks the stored data by satellite. The acoustic and radio links are separate systems that share one memory. The acoustic signal at 25 kHz is demodulated into bits and stored in memory. Hours later, those same bits are modulated onto a 1.6 GHz carrier for satellite transmission. There is no direct transduction between the acoustic and radio signals, and the design does not claim or rely on one.

B. Receiver geometry

We chose a vertical array because the two multipath arrivals we need to distinguish are separated in elevation. The receiver uses a 1 × 24 vertical line array operating at 25 kHz. With half-wavelength spacing, the elements are 3.00 cm apart, giving a 0.69 m aperture, a 4.23° beamwidth and 13.80 dB of broadside directivity. When the array is steered, the directivity falls to 11.25 dB [C], [S].

The choice of aspect ratio matters more than the element count. A 4 × 6 planar arrangement using the same 24 elements has only six elements vertically, resulting in 7.78 dB

of directivity and a 16.92° beam [S]. That beam is too wide to resolve separations of roughly 19°, and the margin only worsens at longer ranges where the two arrivals converge further. The same hardware count therefore succeeds or fails depending on how it is distributed, which is a design conclusion rather than a cost one.

The capsule itself is a slender vertical body measuring 0.20 m by 1.19 m, with a displacement of 37.4 L and a mass of 38.4 kg [C]. Its length is determined principally by the array aperture, and the capsule can be deployed by two people from a small vessel. At this scale it belongs to the same class as existing profiling floats and gliders, which bears on deployment cost.

C. Buoyancy engine and duty cycle

The capsule's vertical motion is controlled by a ball-screw-driven rigid piston [D]. The piston provides a 250 mL stroke, and the mechanism must deliver a peak thrust of 5.70 kN at a depth of 800 m. The ideal work per cycle is $P \cdot \Delta V$ = 1,240 J at 80.6 bar [C]. At a drivetrain efficiency of 0.505 the energy required is 2,455 J, or 0.68 Wh per cycle [C]. An oil-hydraulic bladder at an efficiency of 0.42 would require 2,952 J, a penalty of 16.8% [C]. THK report a positive efficiency of 0.90 for a ball screw [5]; the remaining drivetrain terms are estimates, and the 16.8% margin depends on them.

The direction of that comparison is more robust than its magnitude. A ball screw's dominant loss is rolling friction, which is approximately flat with pressure, whereas a pump's dominant loss is internal leakage, which grows with differential pressure. At 800 m the ball screw is favoured. Peak thrust scales as $P \cdot A$, so at the 2,000 m typical of standard profiling floats the balance moves toward oil-bladder hydraulics, which is why floats at that depth use them. As an independent check, Deep SOLO reports 4.2 kJ for a 2,000 dbar cycle [6]; scaled to 800 m this gives 1.68 kJ against our 2.46 kJ, the same order and conservative in the safe direction. Deep-profiling floats of this class already operate routinely to depths of 6,000 m [31].

At 800 m, with a vertical speed of 0.15 m/s, a complete cycle takes 4 h 13 min [C]. Of this, 1 h 29 min is spent descending, 45 min on the acoustic session, another 1 h 29 min ascending, and 30 min in the surface window. The design cadence is one cycle per day [D], so the capsule is at the surface only 2% of the time, compared with 100% for a moored buoy, and the buoyancy engine consumes approximately 250 Wh per year.

IV. Method

A. Environment

All environmental inputs are measured data. No sound-speed profile, bathymetric value, wave statistic or seabed parameter used in this work was assumed, interpolated from a textbook, or chosen for convenience. We state this explicitly because the central results are propagation results, and a propagation result computed on an invented ocean is not evidence of anything.

Sound speeds were obtained from the NOAA World Ocean Atlas 2023 one-degree statistical means at 49.5° N, 128.5° W [7], converted by the Mackenzie nine-term equation [8], with standard ocean-acoustic relations taken from Urick [24]. The resulting structure is given in Table I. In summer the surface layer is faster than the water at the seabed; in winter it is slower. This inversion is not incidental, and it drives the seasonal behaviour reported in Section VI-B.

TABLE I. Sound-speed structure at the deployment site, from WOA23 [V].

Season	c(0) [m/s]	Channel axis	c(2372) [m/s]	Surface relative to seabed
Summer	1501.33	400 m	1495.81	faster
Winter	1479.81	500 m	1495.61	slower

Bathymetry was taken from the GEBCO 2026 grid [9], which gives 2,341 m at the site with a mean slope of 0.2 m/km over ±36 km. This independently confirms Frye’s reported water depth to within 31 m from a source unconnected to his paper, and it justifies the

range-independent propagation model rather than merely assuming it.

Sea state was derived from Fisheries and Oceans Canada station C46206 on La Perouse Bank [10], using 203,885 quality-controlled records spanning 1988 to 2022. Significant wave heights are 1.95 m at the median, 3.73 m at the 90th percentile, 4.39 m at the 95th and 5.73 m at the 99th. Wave orbital amplitude decays with depth as $\exp(-2\pi z/L)$. Requiring that displacement fall below $\lambda/10$ at 25 kHz, so that the array can hold phase coherence across its aperture, places the beamforming floor at 329 m at the 95th-percentile sea state [C]. This is the lower bound of the park-depth window analysed in Section VII-A.

Seabed properties are Hamilton's geoacoustic values for silt: a density of 1,767 kg/m^3, a compressional speed of 1,623 m/s and an attenuation of 0.673 dB/kHz [11].

B. Propagation model

The underwater acoustic channel varies with range, frequency and depth, with multipath propagation and limited bandwidth being its defining constraints [25]. We model propagation using the Bellhop Gaussian beam tracer [12], which returns the amplitude, delay, source angle, receiver-side arrival angle and bounce counts for each eigenray.

The receiver-side arrival angle is particularly important because each ray can reach the receiver from a different direction. The array's beam pattern must therefore be evaluated at the actual arrival angle of each ray, rather than at one assumed direction. This allows directivity and multipath rejection to be accounted for through the geometry of the propagation paths and the array, rather than being added to the link budget as a constant. It also explains why a 60° transducer and a 4.23° array produce different results even when the ray paths are identical. A model that adds directivity as a scalar gain cannot reproduce the effect reported here, because in such a model both arrivals receive the same gain by construction.

We calculate transmission loss by coherently summing the gain-weighted arrivals and averaging the result over 25 samples across the 4 kHz band. Band averaging is necessary

because a single-frequency coherent sum produces deep interference fringes that are an artefact of the evaluation frequency rather than a property of the channel. Thorpe absorption is applied to each arrival over its actual path length, not over the horizontal range, since the two differ substantially for steep paths in deep water.

C. Two independent tests

Every result in this paper rests on two tests that we keep distinct throughout. The first is geometric: does a direct ray arrive at all? If no direct eigenray connects source and receiver, the link is shadowed, and Zhao *et al.* establish that this condition is not recoverable by raising source level [4]. The second is the link budget: is the arriving ray strong enough to carry data? This condition is recoverable, by power, by coding, or by a longer integration time.

Conflating the two obscures which constraint binds at a given geometry, and the distinction is what makes the coverage result interpretable. A link may possess a path and still fail, or possess no path at all, and the engineering responses to those two situations are entirely different.

D. Validation

The ray engine was validated against the exact closed-form solution for straight rays in an isovelocity ocean, in which the geometry is analytic. For a source at 200 m and a receiver at 2,371 m at 3 km horizontal range, the analytic direct path length is 3,703.1 m arriving at +35.892°. Bellhop returns 3,703.1 m at +35.892°, an error of 0.0000° and 0.00 m [S].

A second and stronger validation appears in the results themselves. The model reproduces the published operating range of two independently deployed systems without being fitted to either. Frye’s link fails beyond 4 km in our simulation, and his paper reports approximately 4 km [1]. The Send glider degrades beyond 3 km, and its paper reports 3.5–5.5 km as typical [2]. Neither figure was used in constructing the model, and no parameter was adjusted to produce them.

V. Results: Multipath Discrimination

A seafloor node transmits once, but its signal reaches the relay twice, once directly and once after reflecting from the sea surface. The copies arrive milliseconds apart at comparable amplitude and interfere, corrupting packets. Measured delay profiles in shallow water confirm surface-reflected arrivals at amplitudes comparable to the direct path [26], so this is an observed effect and not only a modelled one.

The two copies arrive from different directions, and the angular separation between them is the quantity that determines whether a receiver can distinguish them. Table II gives that separation together with the suppression each receiver achieves, and Fig. 1 shows the same comparison against both beamwidths.

TABLE II. Direct-to-surface angular separation and the suppression achieved by each receiver [S]. Both arrivals lie within the main lobe of a 60° beam, whose −3 dB points are at ±30°.

Range	Separation	Moored 60° beam	Array 4.23° beam
1.0 km	15.04°	< 3 dB	13.6 dB
2.0 km	19.76°	< 3 dB	22.5 dB
3.0 km	19.18°	< 3 dB	24.0 dB
4.0 km	17.28°	< 3 dB	26.5 dB
5.5 km	14.47°	< 3 dB	23.6 dB
7.0 km	12.32°	< 3 dB	17.7 dB
10.0 km	9.53°	< 3 dB	32.9 dB
20.0 km	0.21°	—	0.0 dB

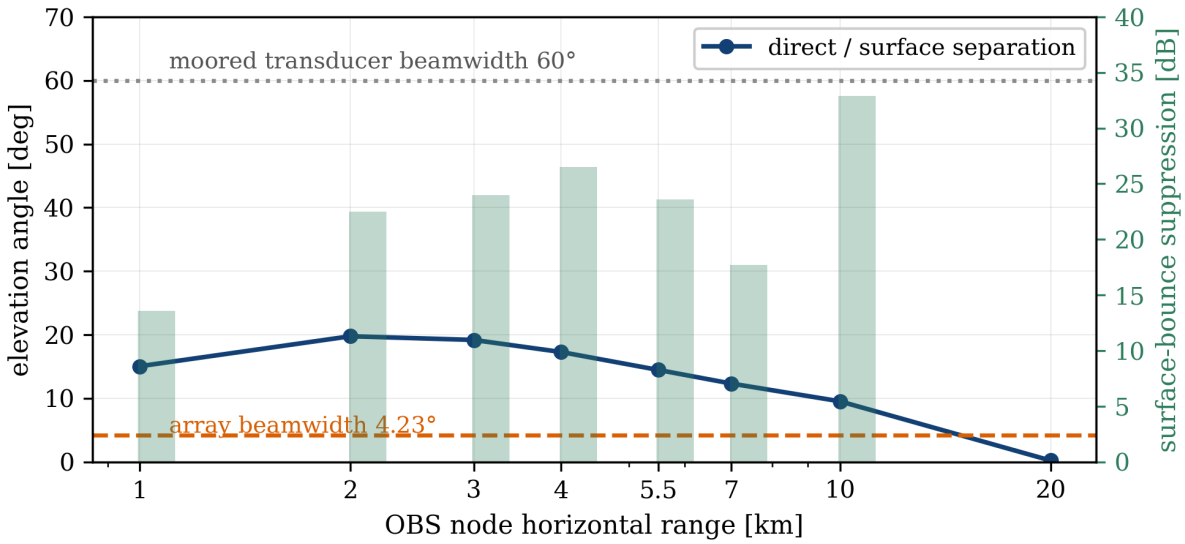


Fig. 1. Angular separation between the direct arrival and its surface reflection, compared against the two receiver beamwidths. The separation lies well inside the 60° main lobe at every range, so the moored transducer cannot discriminate between the arrivals. It remains above the 4.23° array beamwidth out to approximately 15 km. Bars show the suppression achieved by the array on the right-hand axis.

The suppression achieved is 13.6 to 32.9 dB across the operating range, against less than 3 dB for the moored transducer. The moored figure is not zero: a real 60° transducer has a gradual roll-off, and an arrival 20° off boresight is attenuated slightly. But a 60° full-width beam has its −3 dB points at ±30°, so no realistic pattern attenuates an arrival 20° off axis by more than a few decibels. We quote the figure as a bound rather than a value.

Two properties of this result deserve emphasis because they bear on its generality. Winter agrees with summer to within 0.3 dB at every range [S]. The separation is set by the receiver's height above the seabed and its distance from its own surface image, and neither quantity is seasonal. The mechanism is likewise frequency-independent: at half-wavelength spacing, 24 elements give a 4.23° beam at any frequency, with only the physical aperture changing. We confirmed this by repeating the computation at 12, 15, 20 and 25 kHz, obtaining the same suppression column.

The mechanism has a geometric ceiling, which we state rather than leave to be discovered. At 20 km the two paths converge to 0.21° apart, well below the 4.23° beamwidth.

Separating them would require roughly 500 elements and a 15 m aperture, which does not fit on a capsule of this class. The mechanism is therefore bounded at approximately 12 to 15 km regardless of transmit power, and the architecture targets dense local instrument clusters rather than widely distributed arrays.

VI. Results: Coverage and Energy

A. Usable radius

Taking a 10% packet error rate as the usability threshold, the usable radius for each configuration is given in Table III. Fig. 2 shows the underlying packet error rate against range for both seasons.

TABLE III. Usable radius by configuration and season, at a 10% packet error rate threshold [S].

Configuration	Summer	Winter	Seasonal swing
Moored relay, 10 m, 60°	3.22 km	4.51 km	40%
Array alone, fixed at 800 m	3.94 km	3.07 km	28%
Array, depth-agile	5.22 km	5.21 km	0.2%

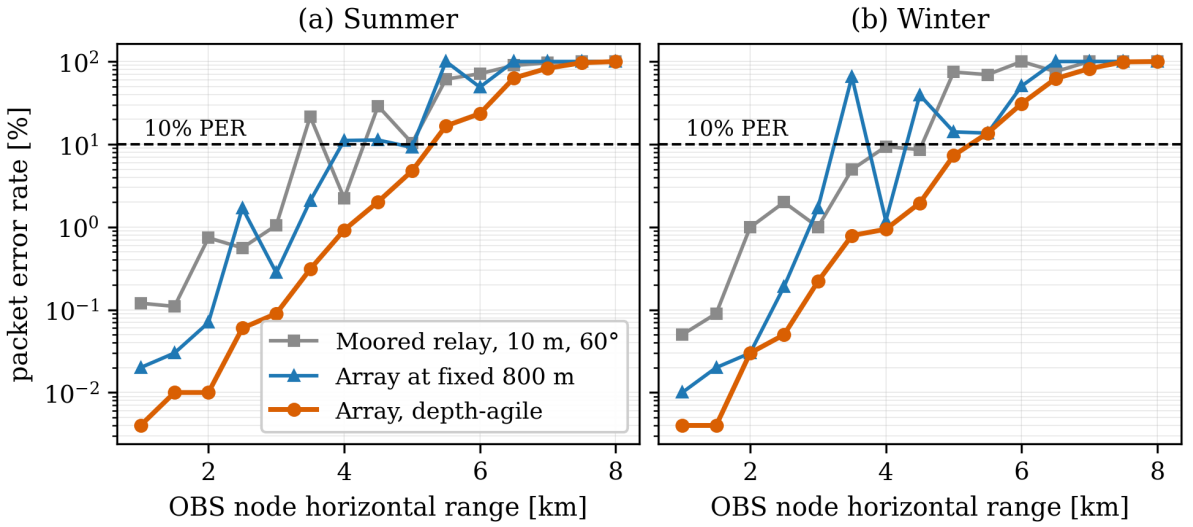


Fig. 2. Packet error rate against node horizontal range for the three configurations, in both seasons. The depth-agile array remains below the 10% usability threshold to 5.22 km in summer and 5.21 km in winter. The moored baseline reaches 3.22 km and 4.51 km respectively. The fixed-depth array falls below the moored baseline in winter.

The depth-agile radius does not move with season. The moored baseline varies by 40% because its transducer is fixed at 10 m while the surface layer is not, and the winter profile is the more favourable of the two for a near-surface source. A relay that re-solves its depth on each cycle lands within 0.01 km of the same radius in both seasons.

This has a direct planning consequence. A moored field must be sized for its worst season, and is therefore over-provisioned for the remainder of the year. A capsule field is sized once. Coverage area scales with the square of radius: 32.5 to 85.6 km^2 in summer, a factor of 2.63, and 63.9 to 85.3 km^2 in winter, a factor of 1.33.

The 3.22 km baseline figure is itself a validation of the model. Frye reports approximately 4 km of telemetry and 5 to 20 km^2 of coverage [1]. We obtain 3.22 km and 32.5 km^2 from physics, with nothing tuned to match, which places the model in the right regime from an independent direction.

B. Energy per delivered byte

Delivering a given payload requires retransmitting failed packets, so packet error rate translates directly into energy. Using the baseline's own deployment parameters, 500 kB per day for 395 days giving 193 MB in total [1], the energy costs are given in Table IV.

TABLE IV. Energy required to deliver a 193 MB payload over a 13-month deployment [S].

Case	Moored relay	Depth-agile array	Saving
5 km, summer	1,804.6 kJ	1,698.8 kJ	5.9%
6 km, summer	5,625.7 kJ	2,113.6 kJ	62.4%
4 km, winter	1,785.4 kJ	1,633.2 kJ	8.5%
5 km, winter	6,384.6 kJ	1,746.1 kJ	72.7%
6 km, winter	no link	2,341.6 kJ	—

The mechanism is the retransmission count: 3.48 against 1.31 transmissions per packet at 6 km in summer, and 3.95 against 1.08 at 5 km in winter [S].

This fractional saving is independent of modem efficiency, which is worth establishing analytically because it removes a parameter that would otherwise require justification. Energy is bits × (transmissions per packet) / η, so in the ratio

$$saving = 1 - (tx_{array} / tx_{moored})$$

the efficiency η appears in both numerator and denominator and cancels exactly. We verified this numerically at 500, 1,000, 1,500 and 2,000 bits per joule, obtaining 62.4% in every case to four significant figures. Only the absolute joule figures scale with η.

We state the limitation of this result plainly. Within 4 km the saving is under 2% and is operationally negligible, because both configurations already deliver most packets on the first attempt. The gain lives at the edge of the neighbourhood, where the baseline's retransmission count climbs steeply. A reader interested only in the 1 to 3 km regime occupied by the reference deployment's own instruments should not expect an energy benefit from this architecture.

C. The satellite uplink

The capsule surfaces to uplink, and the antenna it carries is constrained by the hull rather than by the link budget. At 1.616 GHz over a 1,400 km slant range with 7 W of transmit power and a −10 dB/K satellite G/T, free-space path loss is 159.54 dB. A +2 dBi quarter-wave monopole then yields E_b/N_0 = 35.71 dB against a QPSK requirement of 10.5 dB for a bit error rate of 10^{-6} [C]. The margin is therefore 25.21 dB, and the antenna may be as poor as −23.2 dBi while still closing the link. We are not aware of a prior published statement of this bound for a surfacing acoustic relay.

The bound matters because it admits a conformal element with no coaxial penetration of the pressure boundary, in place of a protruding whip. Magnetoelectric antennas are one candidate class. They resonate mechanically rather than electrically, with $f_0 = (1/2\pi)\sqrt{(k_0/m_0)}$ [13], in which the surrounding medium does not appear, so the resonant frequency does not

shift when the element is wetted. A conventional resonant element sets its frequency by electrical length and shifts by √80 between dry and seawater conditions. Reported devices are conformal at approximately 1% of the electromagnetic wavelength [14], and have been demonstrated at MEMS scale [27] and on underwater robot-to-robot links [28]. We report the bound as the result and the antenna technology as a candidate that satisfies it, not as a contribution of this work.

D. The reverse link

The acoustic link is bidirectional in operation, since the capsule must deliver commands, clock corrections and retransmission requests to the instrument, as the reference glider system also reports [2]. An aperture is reciprocal, so the same 24 elements form a 4.23° transmit beam with the same 13.80 dB of directivity, and two consequences follow.

First, concentrating that gain toward the instrument is a power ratio of 24×, so the same received level at the seafloor requires a source level of 176.2 dB rather than 190 dB, approximately 0.33 W in place of 8 W. On a platform executing one cycle per day for a year, this places the acoustic downlink well below the 0.68 Wh consumed per buoyancy cycle.

Second, where the forward link rejects a reflection that already exists, a transmit beam that does not illuminate the sea surface never creates one. The seafloor instrument listens with a single omnidirectional hydrophone and cannot discriminate spatially, so the array is the only element in the link able to improve the downlink at all. This is a stronger statement than the forward-link result, because the instrument itself cannot be modified without recovering it. Both consequences follow from reciprocity and the beam pattern of Table II rather than from separate simulation, and we present them as such.

VII. Discussion

A. The park-depth window

Park depth is bounded on both sides by measured physics, and the upper bound is not fixed. The lower bound is 329 m, set by the 95th-percentile sea state [10]; below it the array cannot hold phase coherence against wave orbital motion. Profile inversion from in-situ measurement is an active field [30], and the capsule derives its park depth from a cast rather than from climatology.

The upper bound is geometric reach, which falls as the relay descends. Water below the sound-channel axis is faster, so a deeper source requires a steeper launch angle to reach a seabed target, and beyond a certain depth no ray connects the two at a given range. Table V gives the relationship and Fig. 3 shows the resulting window.

TABLE V. Maximum geometric reach to a seabed node as a function of relay park depth, summer profile [C].

Park depth	**400 m**	**800 m**	**1,200 m**	**1,600 m**	**2,000 m**
Maximum reach	20.1 km	17.9 km	15.1 km	12.0 km	8.3 km

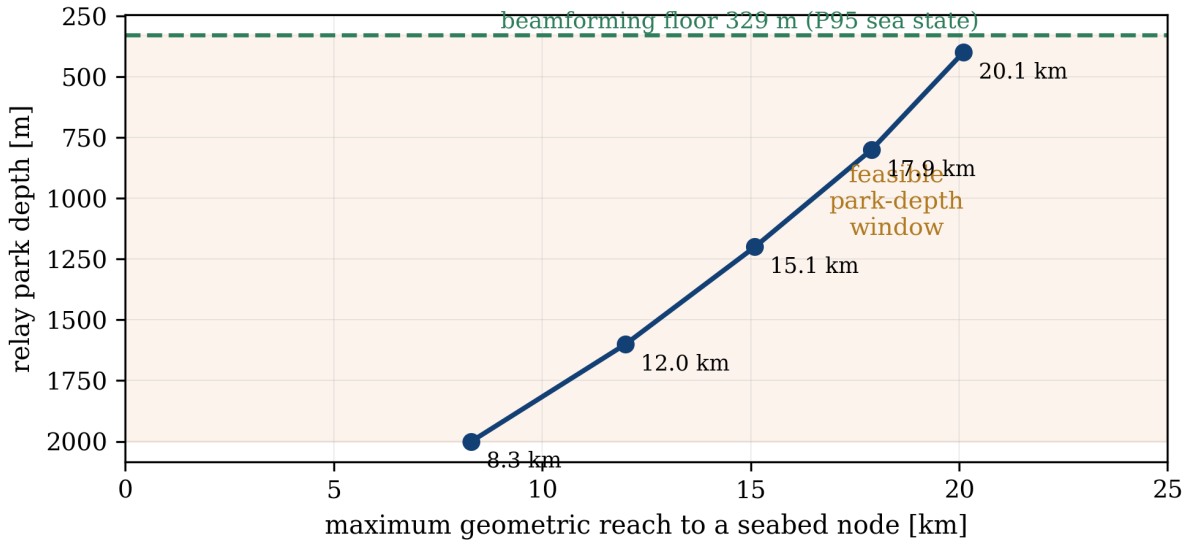


Fig. 3. The park-depth window. The lower bound at 329 m is set by the 95th-percentile sea state, below which the array cannot hold phase against wave orbital motion. The upper bound is set by geometric reach, which falls from 20.1 km at 400 m to 8.3 km at 2,000 m, so the window contracts as the target range increases.

Shadow-zone extent is conventionally established by ray simulation [32], and two independent methods agree on this boundary. A Snell ray integration predicts that a relay at 2,000 m cannot reach a node at 10 km, and the Bellhop tracer returns exactly that cell as shadowed in a 70-cell grid [S]. The agreement is a useful internal check, since the two methods share no code.

The consequence is that the ceiling moves with target range: approximately 2,000 m for a node at 8 km, and approximately 450 m for a node at 20 km. No single fixed depth is simultaneously optimal for a field containing instruments at different ranges, which is the physical basis for depth agility and the reason a fixed installation cannot be pre-positioned to serve them all.

We note that park depth is not the sound-channel axis. The axis lies at 400 to 500 m and is the correct criterion when both endpoints are in mid-water. For a seabed target what matters instead is the receiver's distance from its own surface image, since that distance sets the multipath separation reported in Section V. These two criteria give different answers, and using the mid-water one for a seabed link is a straightforward way to arrive at the wrong depth.

B. Neither element is sufficient alone

A natural objection to this architecture is that one might simply mount an array on the existing mooring, obtaining the angular resolution without the mechanical complexity of a buoyancy engine. Table III answers it directly. The same 24 elements held at a fixed 800 m reach 3.94 km in summer and 3.07 km in winter, and the winter figure is below the moored baseline's 4.51 km.

An array at an unsuitable depth is therefore worse than no array at all. Angular resolution and the ability to position that resolution are each insufficient in isolation, and the result follows from the combination rather than from either element. For the same reason, this comparison is the closest available analogue to a stationary-versus-mobile evaluation within a

two-point geometry, and it serves the purpose for which [4] would otherwise be the natural baseline.

C. Scope and applicability

Radius is not a data-rate result, and we are explicit about this because the distinction is easy to lose. The total data recoverable from any single instrument is unchanged: a relay at 6 km moves the same bits per second as one at 3 km. What the radius determines is the geometry a deployment can adopt.

Instrument spacing in acoustically telemetered deployments is constrained by relay reach rather than chosen independently of it. The reference deployment placed its instruments within 3 km against a stated 4 km telemetry limit [1]; the array was laid out to fit the relay, not the reverse. Published spacings span a wide range by experiment type. Academic wide-azimuth refraction profiles typically use 5 to 10 km receiver spacing, cabled seafloor systems have been deployed at 5 km, and mantle-scale passive arrays such as UPFLOW use 110 to 160 km [18]. A usable radius of 5.22 km brings the refraction-profile regime within reach of a single relay position and relaxes the constraint on array design in the denser regimes. At mantle scale no acoustic relay of any kind is applicable, and physical recovery remains the only option.

We do not claim that a drifting relay substitutes for a propelled vehicle. Where instruments are widely scattered, mobility is the dominant variable and an AUV data mule is the appropriate architecture [15], [16]. The results reported here are array-side and transfer to that architecture unchanged, since Sections V and VI-B concern the receiver itself rather than the platform carrying it. The capsule embodiment is advantageous specifically where a persistent relay is preferable to repeated vehicle transits: long deployments, clustered instruments, and programmes in which each vehicle mission carries a vessel cost.

VIII. Limitations and Assumptions

Three assumptions [A] remain in the acoustic model, and we enumerate them so that a reader can judge their effect.

WOA23 terminates at 2,200 m at this grid cell, so the final 172 m of the water column is linearly extrapolated in temperature and salinity. This is under 16% of the column, in near-isovelocity water, and the extrapolated segment contains no feature that would alter the ray geometry materially.

The 60° baseline transducer is modelled with a flat top and a −20 dB skirt. Real patterns roll off gradually, which is why Table II reports a bound of less than 3 dB rather than a specific value. The conclusion is unaffected, since no realistic 60° pattern rejects an arrival 20° off boresight by more than a few decibels, but the exact figure is a modelling artefact and should not be quoted as a measurement.

Drivetrain efficiencies other than the THK ball-screw figure [5] are estimates. The 16.8% buoyancy margin in Section III-C depends on them, although the direction of the comparison does not. The radio-frequency budget in Section VI-C uses standard Iridium-class link values [A].

Four limitations of method also apply. Bellhop is a two-dimensional ray model over a horizontally uniform ocean, and does not represent currents, internal waves, surface or seabed roughness, or out-of-plane propagation. This is standard practice in the underwater acoustic literature, and the measured 0.2 m/km slope [9] supports the range-independent approximation at this particular site, but it remains an approximation.

The Rayleigh fading model requires 33.9 dB SNR for a 10% packet error rate. Beamforming partly creates a Rician channel, which would be more favourable and is not modelled, so the figures reported here are conservative in the direction that matters.

The array is steered exactly at the direct arrival. A real system estimates that angle with error, and on a 4.23° beam a pointing error of one or two degrees costs several decibels of main-lobe gain and displaces the null away from the surface arrival. Adaptive weighting, by MVDR or LCMV, is the standard mitigation and is the subject of ongoing work. We regard

this as the most significant gap between the simulation and a deployable system.

Finally, the entire study is simulation. The ray engine is validated against a closed form and the model reproduces two published operating ranges, but no tank or sea trial has been conducted. The suppression figures and the 5.22 km radius should be read as modelled quantities pending experimental confirmation.

IX. Conclusion

The reach of an acoustically linked seafloor observatory is limited by an interference mechanism that transmit power cannot address. At the Nootka fault the direct path and its surface reflection arrive between 9.5° and 19.8° apart across 1 to 10 km, and the 60° transducer of a deployed moored observatory suppresses the reflection by less than 3 dB. A 24-element vertical line array with a 4.23° beamwidth suppresses it by 13.6 to 32.9 dB, varying by under 0.3 dB between seasons and unchanged across the 12 to 25 kHz band.

Because such an array must be mechanically decoupled from surface wave motion, with a floor of 329 m at the 95th-percentile sea state, it is carried on a buoyancy capsule that selects its park depth on each dive. Computed on measured oceanography, this extends the usable radius from 3.22 km to 5.22 km in summer and from 4.51 km to 5.21 km in winter, with the depth-agile radius varying by 0.2% against 40% for the moored baseline. The same 193 MB payload costs 62.4% less energy at the edge of that neighbourhood, a fraction shown analytically to be independent of modem efficiency. The satellite uplink carries 25.21 dB of margin, admitting a conformal antenna of −23.2 dBi and therefore a hull with no coaxial penetration.

An ablation establishes that neither element suffices alone: the same array held at a fixed 800 m reaches only 3.07 km in winter, below the moored baseline. The contribution is array-side and platform-independent. Multipath rejection, retransmission cost and reverse-link focusing hold on any platform capable of carrying the array, and in the formulation used by AUV path-planning work [15], [16] the receiver enlarges the

communication neighbourhood those methods take as a fixed input.

Future work will replace the climatological sound-speed profile with a measured cast at the deployment site, implement adaptive beamforming to make the suppression robust against pointing error, evaluate the architecture against a surveyed instrument geometry rather than a circular coverage model, and conduct tank-scale validation of the array response before any sea trial.

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